Lessons: the gateway and serving layer

Every failure was logged, then fixed or explained

what happened | fixed during the experiment | future work

A replica re-added to the pool looked dead (503 no_signal)	one object per replica URL; scrape before answering
Placement remembered a cache that had been emptied	a cache reset also clears the gateway's memory
Prefix-aware v1: biased tie-break, oversized memory	v2: least-busy matching replica, memory sized (+36%)
A dead replica kept getting half the calls	route only to replicas answering their metrics (188 → 8)
Admission locked itself out after preemption	probe calls, decay the learned rate, vLLM's prefill time